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EXPERIMENT - 15

Multi-Agent Reinforcement Learning and Partially Observable Markov Decision Process (POMDP) Simulation

Aim

To implement a simple Multi-Agent Reinforcement Learning environment and simulate a Partially Observable Markov Decision Process (POMDP) where multiple agents interact and learn through rewards.

Procedure

1. Create a simple grid environment.
2. Define the positions of two agents and the goal.
3. Allow both agents to select actions.
4. Provide limited observations to simulate partial observability.
5. Calculate rewards based on the agents' movements.
6. Move the agents according to their selected actions.
7. Repeat the simulation for multiple steps.
8. Record the rewards obtained by both agents.
9. Visualize the agents' rewards and positions.

Code:

```
# =====  
  
# EXPERIMENT 15  
  
# MULTI-AGENT RL AND POMDP SIMULATION  
  
# =====  
  
import numpy as np  
import matplotlib.pyplot as plt  
  
np.random.seed(42)  
  
# -----
```

```

# Environment
# -----

grid_size = 5
goal = np.array([4, 4])

# Two agents
agents = [
    np.array([0, 0]),
    np.array([0, 1])
]

# Actions: Up, Down, Left, Right
actions = [
    np.array([-1, 0]),
    np.array([1, 0]),
    np.array([0, -1]),
    np.array([0, 1])
]

reward_history = [[], []]
position_history = [[], []]

# -----
# Movement Function
# -----

def move(position, action):

    new_position = position + action

    # Keep agent inside grid
    new_position = np.clip(
        new_position,
        0,

```

```

        grid_size - 1
    )

    return new_position

# -----
# Simulation
# -----

for step in range(50):

    for i in range(2):

        # Partial observation:
        # Agent observes only its own position
        observation = agents[i].copy()

        # Simple random policy
        action_index = np.random.randint(
            len(actions)
        )

        agents[i] = move(
            agents[i],
            actions[action_index]
        )

        # Reward
        if np.array_equal(
            agents[i],
            goal
        ):
            reward = 10
        else:

```

```

        reward = -1

    # Store results
    reward_history[i].append(reward)

    position_history[i].append(
        agents[i].copy()
    )

# -----
# Results
# -----

print("=" * 55)
print("MULTI-AGENT RL AND POMDP RESULTS")
print("=" * 55)

for i in range(2):

    print(
        f"\nAgent {i + 1}"
    )

    print(
        "Total Reward:",
        sum(reward_history[i])
    )

    print(
        "Final Position:",
        agents[i]
    )

# -----
# Visualization 1: Reward Comparison

```

```
# -----
```

```
plt.figure(figsize=(10, 5))
```

```
plt.plot(  
    reward_history[0],  
    label="Agent 1"  
)
```

```
plt.plot(  
    reward_history[1],  
    label="Agent 2"  
)
```

```
plt.title(  
    "Multi-Agent Reward Comparison"  
)
```

```
plt.xlabel("Step")  
plt.ylabel("Reward")
```

```
plt.legend()  
plt.grid(True)
```

```
plt.show()
```

```
# -----
```

```
# Visualization 2: Agent Positions
```

```
# -----
```

```
positions1 = np.array(  
    position_history[0]  
)
```

```
positions2 = np.array(  
    position_history[1]  
)
```

```

        position_history[1]
    )

plt.figure(figsize=(7, 7))

plt.plot(
    positions1[:, 1],
    positions1[:, 0],
    marker="o",
    markersize=3,
    label="Agent 1"
)

plt.plot(
    positions2[:, 1],
    positions2[:, 0],
    marker="o",
    markersize=3,
    label="Agent 2"
)

plt.scatter(
    goal[1],
    goal[0],
    marker="*",
    s=200,
    label="Goal"
)

plt.xlim(-0.5, grid_size - 0.5)
plt.ylim(-0.5, grid_size - 0.5)

plt.xticks(range(grid_size))
plt.yticks(range(grid_size))

```

```
plt.xlabel("Grid X")
plt.ylabel("Grid Y")

plt.title(
    "Multi-Agent POMDP Grid Simulation"
)

plt.legend()
plt.grid(True)

plt.show()
```

OUTPUT:-

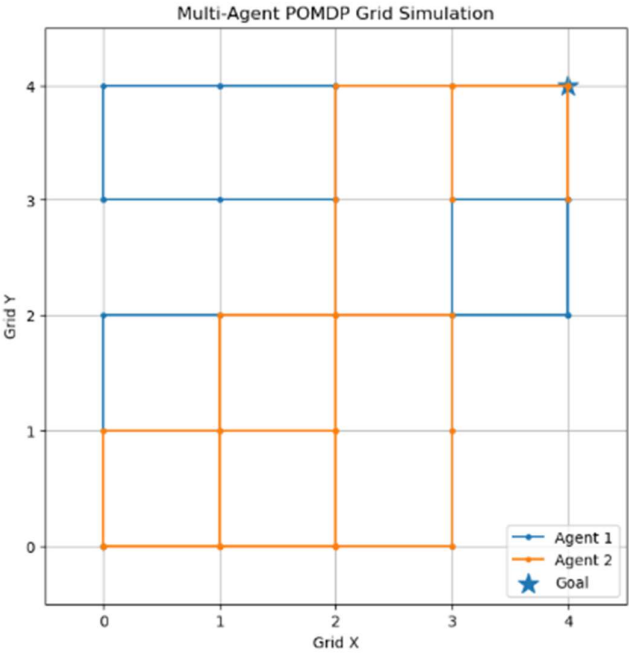
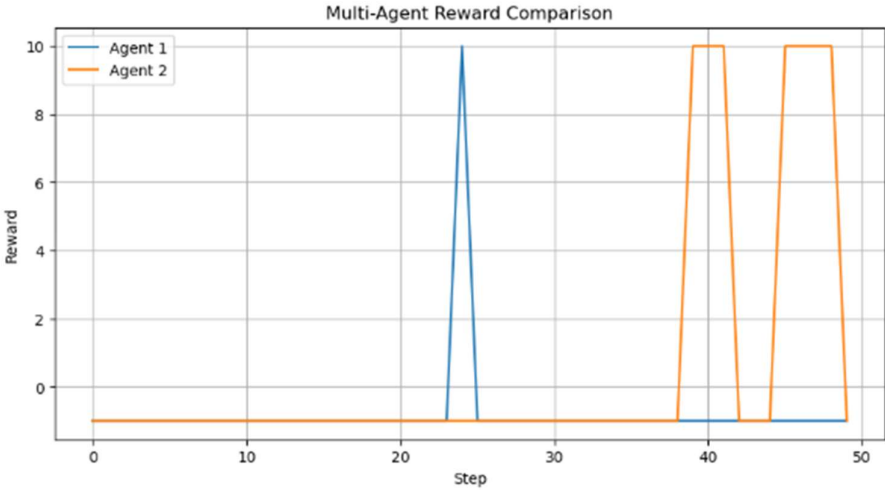
=====

MULTI-AGENT RL AND POMDP RESULTS

=====

Agent 1
Total Reward: -39
Final Position: [4 2]

Agent 2
Total Reward: 27
Final Position: [3 4]



Visualization

The program produces two visualizations:

1. Multi-Agent Reward Comparison
Shows the rewards received by Agent 1 and Agent 2 during the simulation.
2. Multi-Agent POMDP Grid Simulation
Shows the movement paths of both agents and the target goal position on the grid.

Summary

A simple Multi-Agent Reinforcement Learning and POMDP simulation was implemented using two agents. Each agent receives only limited information about the environment, representing partial observability, and interacts with the environment through actions and rewards.

Result

Thus, Multi-Agent Reinforcement Learning and POMDP simulation were successfully implemented, and the agents' rewards and movement paths were visualized.